Final Exam

Graded

Student

Aris Rault

Total Points

40 / 45 pts

Question 1

Overall Grades **40** / 45 pts

✔ **– 0 pts** See the details on each problem.

– 5 pts Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Aris Rault	ar846	

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: 

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	4	5	5	5	5	5	5	5	1	40

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Problem 1. [5'] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

augmented matrix form:

$$\Rightarrow \left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 2 & -3 & 0 & 1 \end{array}\right] \xrightarrow[\sim]{R_3 \leftarrow R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -7 & 1 \\ 0 & -3 & -4 & -1 \end{array}\right]$$

$$\xrightarrow[\sim]{R_3 \leftarrow -R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 0 & 3 & 4 & 1 \end{array}\right] \xrightarrow[R_3 \leftarrow -R_3]{2R_3 \leftarrow 2R_3 + 3R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right] \xrightarrow[\sim]{\substack{R_2 \leftarrow R_2 + 3R_3 \\ R_1 \leftarrow R_1 - 2R_3}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & -2 & 0 & 4 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

This is why you obtained the wrong number

$$\xrightarrow[\sim]{R_2 \leftarrow -\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$(x, y, z) = (-1, -2, 1)$ is the general and unique solution.

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Problem 2. [5] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(1) and (2)

$A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ $R_1 \leftarrow \frac{1}{2}R_1$, $R_2 \leftarrow \frac{1}{5}R_2$ $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ $R_2 \leftarrow R_2 + R_1$, $R_4 \leftarrow R_4 - R_1$, $R_2 \leftarrow \frac{1}{2}R_2$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ $R_3 \leftarrow R_3 - R_2$, $R_4 \leftarrow R_4 - R_2$ $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ pivots on column 1, 2 // // row 1, 2

column space $= \left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

row space $= \left\{ \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix} \right\}$

(3)

$Av = 0 \iff \left[\begin{array}{ccc|c} 2 & 4 & 2 & 0 \\ -5 & 0 & 5 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \end{array}\right]$ same steps $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ $x = -2y - t \to 2t - t = t$, $y = -t$, $z = t$

so $t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ so null space $\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

We want formula for T, $A = SB^{-1}$ $\quad S = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$

$$B = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$$

$[B\,|\,I] \Rightarrow \left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\-1&-1&-1&0&0&1\end{array}\right]$ $R_3 \leftarrow -R_3$ $\left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\1&1&1&0&0&-1\end{array}\right]$ $R_2 \leftarrow R_1$

$\left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\1&1&1&0&0&-1\end{array}\right]$ $R_3 \leftarrow R_3 - R_1$, $R_3 \leftarrow R_3 - R_2$ $\left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&1&-1&-1&-1\end{array}\right] = [I\,|\,B^{-1}]$

So $B^{-1} = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$

$A = SB^{-1} \Rightarrow \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix} = A.$

Problem 4. [5'] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$

find $p(\lambda)$,

$(A - I\lambda) \Rightarrow \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$ $\det(A - I\lambda) = \det(A) = 0$

So $(2-\lambda)\begin{vmatrix} 1-\lambda & 0 \\ -2 & 1-\lambda \end{vmatrix} = (2-\lambda)\left[(1-\lambda)(1-\lambda) - 0\right] = (2-\lambda)\left(1 - 2\lambda + \lambda^2\right)$

So $p(\lambda) = (2-\lambda)(\lambda-1)^2$, eigenvalues are roots. $\lambda_3 = 2$ $\lambda_1 = \lambda_2 = 1$.

$\lambda_3 = 2$ $\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix} \overset{R_3 \leftarrow R_3 + 2R_1}{\sim} \begin{bmatrix} -1 & 2 & 0 \\ 2 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \overset{R_2 \leftarrow R_2 + 2R_1}{\overset{R_1 \leftarrow R_1 + R_2}{\sim}} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 6 & 2 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 2 & 0 \\ 0 & 6 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $x = -2y = 6x = 2/6 t$, $6y = -t$ $y = -1/6 t$, $z = t$

so $t\begin{bmatrix} 1/3 \\ -1/6 \\ 1 \end{bmatrix}$ for $t = 6$ $V_3 = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$

$\lambda_2 = \lambda_1 = 1$ $\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \overset{R_3 \leftarrow -1/2 R_3}{\overset{R_1 \leftrightarrow 1/2 R_1}{\sim}} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \overset{R_1 \leftarrow R_1 - R_2}{\sim} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $x = 0$, $y = 0$, $z = t$

$V_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ for V_2 $(A - I)V_2 = V_1$ s. $\left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 1 \end{array}\right] \overset{R_1 \leftrightarrow 1/2 R_1}{\sim} \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 1 \end{array}\right] \overset{R_2 \leftarrow R_2 - R_1}{\sim}$

$\left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -2 & -2 & 0 & 1 \end{array}\right] \overset{R_3 \leftrightarrow R_3}{\sim} \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \overset{R_1 \leftrightarrow R_2}{\sim} \left[\begin{array}{ccc|c} -2 & -2 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ $x = -1/2$, $y = 0$, $z = t$ $t\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1/2 \\ 0 \\ 0 \end{bmatrix}$

so for $t = 0$ we have $(A - I)V_2 = V_1$ So $V_2 = \begin{bmatrix} -1/2 \\ 0 \\ 0 \end{bmatrix}$

So we have a block of 2x2 of λ_1 & λ_2 So $J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$P = \begin{bmatrix} 0 & -1/2 & 2 \\ 0 & 0 & -1 \\ 1 & 0 & 6 \end{bmatrix}$ $P^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -4 & 0 \\ 0 & -1 & 0 \end{bmatrix}$

$PJP^{-1} = \begin{bmatrix} 0 & -1/2 & 2 \\ 0 & 0 & -1 \\ 1 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ -2 & -4 & 0 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$

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Problem 5. [5] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$$f(x,y,z) = \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Polynomial → eigenvalues:

$$\det(A - I\lambda) = \det \begin{vmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{vmatrix} = (3-\lambda)\left[(3-\lambda)(4-\lambda) - 1\right] + (3-\lambda) = (3-\lambda)\left[12 - 7\lambda + \lambda^2 - 1 + 1\right]$$

$$\Leftrightarrow \quad p(\lambda) = (3-\lambda)(\lambda^2 - 7\lambda + 13) = (3-\lambda)^2(\lambda - 4) \quad \Leftrightarrow \quad (3-\lambda)(\lambda - 5)(\lambda - 2)$$

$\lambda = 4$

$$(A - 4I) = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 1 & -1 & 0 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_1} \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_3} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_3} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \xrightarrow{\substack{x=0 \\ y=0 \\ z=0}} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$\lambda = 3$

$$A - 3I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} x - y + z = 0 \\ -z = 0 \\ 0 = 0 \end{matrix} \quad \begin{matrix} x = t \\ y = t \\ z = 0 \end{matrix} \quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$(A - 5I) = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ -2 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} x - y - z = 0 \\ 2y + z = 0 \\ 0 = 0 \\ z = t \end{matrix} \quad \begin{matrix} x = -1/2\,t \\ y = -1/2\,t \\ z = t \end{matrix} \quad t = 2 \quad v_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

So $P = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix}$ $\quad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ $\quad P = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix}$ orthogonal $\quad P^{-1} = P^T$ if P is normalized

$\lambda = 2$

$$\text{for } \vec{v}.(A - 2I) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} x - z = 0 \\ y - z = 0 \\ 0 = 0 \\ z = t \end{matrix} \quad v_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

for x, y, z

v_1, v_2, v_3 are orthogonal.

$$P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix} \quad P^{-1} = P^T = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{6} & -1/\sqrt{6} & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$$

we change variables $x \to Pv$ So $x^T A x \to (Pv)^T P x$

So $v^T P^T P P v \Leftrightarrow v^T D v \Leftrightarrow (v,v,w) \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} v \\ v \\ w \end{bmatrix}$

$(v,v,w) =$

So $f(v,v,w) = 3v^2 + 5vv^2 + 2w^2$.

Problem 6. [5'] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

we want $\text{span}\{v_1 v_2 v_3\}$ to become basis.

So casting out

$$\begin{bmatrix} 0 & 2 & -1 \\ 2 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix} \overset{R_3 \leftrightarrow R_1}{\sim} \begin{bmatrix} -2 & 2 & 0 \\ 2 & -1 & 1 \\ 0 & 2 & -1 \end{bmatrix} \overset{R_2 \leftarrow R_2 + R_1}{\sim} \begin{bmatrix} -2 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & -1 \end{bmatrix} \overset{R_3 \leftarrow R_3 - 2R_2}{\sim}$$

$$\begin{bmatrix} -2 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & -3 \end{bmatrix} \overset{\substack{R_1 \leftarrow \frac{1}{2}R_1 \\ R_3 \leftarrow -\frac{1}{3}R_3}}{\sim} \begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \overset{R_1 \leftarrow R_1 - R_2}{\sim} \begin{bmatrix} -1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \overset{\substack{R_1 \leftarrow R_1 - R_3 \\ R_2 \leftarrow R_2 - R_3}}{\sim} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$ so all ind.

$w = av_1 + bv_2 + cv_3$ is the coordinates of Basis $\left\{\begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}\right\}$ so $[w]_B$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \overset{R_3 \leftrightarrow R_1}{\sim} \left[\begin{array}{ccc|c} -2 & 2 & 0 & -8 \\ 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & 3 \end{array}\right] \overset{R_2 \leftarrow R_1 - R_2}{\sim} \left[\begin{array}{ccc|c} -2 & 2 & 0 & -8 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & -1 & 3 \end{array}\right] \overset{\substack{R_1 \leftarrow \frac{1}{2}R_1 \\ R_3 \leftarrow R_3 - 2R_2}}{\sim}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & -3 \end{array}\right] \overset{R_3 \leftarrow -3R_3}{} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right] \overset{R_2 + R_3}{\sim} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right] \overset{R_1 \leftarrow R_1 + R_2}{\sim}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right] \quad \text{so} \quad w_{[B]} = [3 \ -1 \ 1]^T \qquad w = 3v_1 - v_2 + v_3 .$$

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Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

We use gram-schmidt to orthogonalize.

$$\begin{bmatrix}0\\1\\1\\0\end{bmatrix} \cdot \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} = 2 \neq 0 \qquad v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix} \qquad v_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$$

$$u_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = v_1 \qquad u_2 = v_2 - \frac{u_1 \cdot v_2}{\|u_1\|^2} u_1$$

$$u_1 \cdot v_2 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}\begin{bmatrix}3\\0\\2\\-1\end{bmatrix} = 2$$

$$u_1 \cdot u_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}\begin{bmatrix}0\\1\\1\\0\end{bmatrix} = 2$$

$$u_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2}\begin{bmatrix}0\\1\\1\\0\end{bmatrix} \Rightarrow u_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$$

$$u_1 \cdot u_2 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix} \cdot \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix} = 0 \text{ so orthogonal}$$

Now normalize,

$$\begin{bmatrix}0\\1\\1\\0\end{bmatrix} \to \begin{bmatrix}0\\1/\sqrt{2}\\1/\sqrt{2}\\0\end{bmatrix} \qquad \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix} \to \begin{bmatrix}3/\sqrt{12}\\-1/\sqrt{12}\\1/\sqrt{12}\\-1/\sqrt{12}\end{bmatrix}$$ length are all 0.

So orthonormal basis for V is V= basis $\left\{\begin{bmatrix}0\\1/\sqrt{2}\\1/\sqrt{2}\\0\end{bmatrix}, \begin{bmatrix}3/\sqrt{12}\\-1/\sqrt{12}\\1/\sqrt{12}\\-1/\sqrt{12}\end{bmatrix}\right\}$

Problem 8. [5] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

if $A^T = -A$ then A is a skew-symmetric matrix

$A^T = -A \iff A^T A^{-1} = -A A^{-1}$

$A^T A^{-1} = -I$

So

$$\begin{bmatrix} -1 & 0 & \cdots & 0 \\ 0 & -1 & & \\ \vdots & & \ddots & \\ 0 & & & -1 \end{bmatrix}_{nk}$$

if nk is even

$\det(A)$ = product of the main diagonal

$-1^{(2k+1)n}$

$A^T = -A$

$-A^T = (A^T)^T = A = -A^T$

$\det(A)$ products of the main diagonal $\det(A) = \det(-1A^T)$ $\det(A^T) = \det(A)$

$\det(A) = -1^n \det(A)$

~~-1^n if $n = 2k+1$ so odd~~

~~$-1^n \det(A) \iff -1^{2k+1} \det(A) \iff -1^2 \cdot 1^{k+1} \det(A)$~~ $\det(cA) = c^n \det(A)$

Lets test for if n is even

$n = 2k$ $-1^n \det(A) = -1^{2k} \det(A) \neq 0$

$-1^2 1^k \det(A)$ cannot equal zero.

$n = 2k+1$

$-1^{2k+1} \det(A) = -1^{2k+1} \det(A) = \det(-(2k+1)A)$

$= \det(-2kA + A) = \det(2kA^T + A)$

$\det(2kA^T + A) \times \det(-2kA) = 0$

↓ can't be 0, so odd $\det(A) = 0$.

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Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

No, it should be $\vec{u}\cdot\vec{v}=0$

we want $\vec{u}$ $\vec{v}$ so that they are orthogonal so that $\|\vec{u}\|\|\vec{v}\| = 0$

$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$, because both positive, $\|\vec{u}+\vec{v}\|^2 = \|\vec{u}-v\|^2 \Leftrightarrow \|u+v\|^2 - \|u\|-\|v\|^2 = 0$.

for $\vec{u}+\vec{v} = a$ $\quad \vec{u}-\vec{v} = b$

not really

$\|a\|^2 - \|b\|^2 = 0$

$\|a-b\|^2 = 0 \quad \Leftrightarrow \quad a^2 + 2ab - b^2 = 0$

$\Rightarrow a u^2 + 2uv + v^2 + 2\left[(u+v)(u-v)\right] - a(u^2 + 2uv - v^2) = 0$

$a^2 \rightarrow u^2 + 2uv + bv^2 + 2u^2 - 2v^2 - u^2 - 2uv + v^2 = 0$

$2u^2 = 0 \qquad 2 \neq 0 \text{ so } \|u\|^2 = 0 \text{ so } \|u\| = 0.$

So $\|\vec{u}\|\|\vec{u}\| = 0$ so $\vec{u}$ and $\vec{v}$ are perpendicular and so orthogonal.